AI-Based Network Security Threat Detection and Response: *Architecture Design and System Improvement*

Artificial Intelligence for Network Security
and Threat Prevention

By Ayush Kumar

Department of Information Technology
University Roll Number: 12200224004
Class Roll Number: 35
Year: 2nd

St. Thomas College of Engineering and Technology
affiliated to
Maulana Abul Kalam Azad University of Technology, West Bengal

Abstract

Network security in the modern world depends largely on the machine learning based signatures detection systems to identify and prevent cyber threats, while these systems are effective against well-documented attack types, they are still vulnerable to subtle behavior low-frequency attacks such as polymorphic and metamorphic attacks and Advance Encrypted Traffic Attacks. The traditional methods fall critically short in these areas. This limitation creates exploitable gaps in an organization's defenses which could not be addressed by single-model architectures.

To mitigate this, an AI based network security threat detection and response architecture is proposed; it is a refined system modelled around a multi-model ensemble that combines convolutional, temporal, and anomaly-based detection capabilities. The new architecture integrates external threat intelligence, automated confidence-based alert triage, hierarchal response actions, and a retraining feedback loop. This would address all the structural weaknesses of conventional single-model approaches.

The original and proposed architecture has been evaluated on the **CSE-CIC-IDS2018** benchmark dataset. The proposed architecture demonstrates measurable improvements across all performance metrics compared to the baseline system. Criteria like detection accuracy, precision and F1-score indicated notable improvements while the false positive rate is reduced quite significantly. The areas with substantial improvements were in attack categories where the original architecture performed the poorest.

These findings suggest that proceeding beyond the single-model detection towards an intelligent, layered and adaptive AI architecture represents a practical and effective path to stronger network security which can handle both known threats and ever-changing attack landscape that conventional systems often fail to anticipate.

Introduction

The reliance on digital infrastructure across every sector of society has made network security a topic of great interest for organizations across the globe. The growth of cloud computing, remote access systems and interconnected devices, the attack surface has expanded for the threat actors considerably. Threats today are ranging from large-scale volumetric attacks specifically designed to overwhelm the network resources to niche, slow-moving intrusions, likes of a backdoor, engineered to go undetected for extended periods. In either the case, the consequences like data breaches, financial loss, business disruption and reputational harm, are substantial and difficult to recover from, such as the 2022 ransomware attack on the **All India Institute of Medical Sciences (AIIMS)** in New Delhi, which paralyzed critical hospital functions and compromised massive amounts of healthcare data.

To defend against these threats, organizations deploy machine-learning based signature detection and intrusion prevention systems. These systems analyze network traffic and run them against observed behavior of libraries of known attack patterns and flagging them as potential threats. Attacks like **Distributed Denial-of-Service** (DDoS) this approach works quite effectively. However, it is fundamentally constrained by its pre-trained data. It can only flag known threats, when met with subtle, low-frequency or unknown attack categories, such as network infiltration or Heartbleed exploitation. The detection rates fall considerably short, leaving behind blind spots in the organizations' defense.

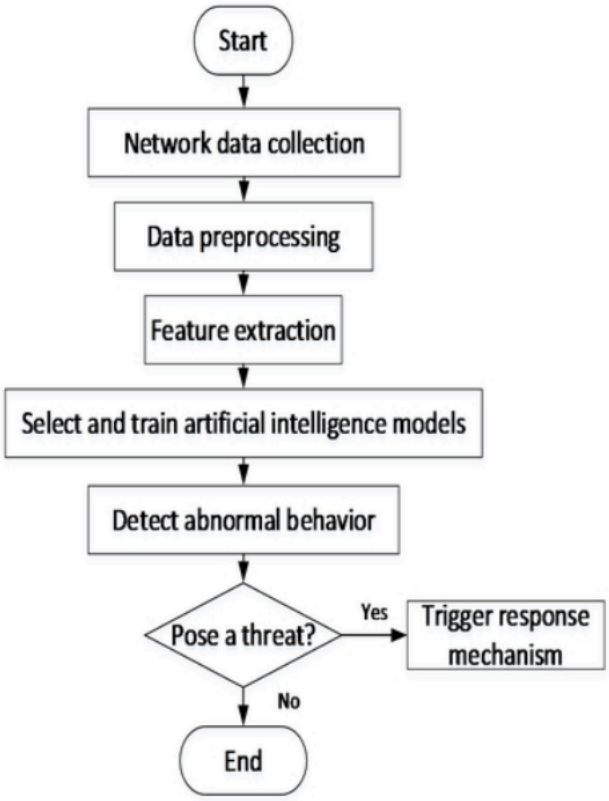
The threat actors are well aware of this limitation. Advanced persistent threats and sophisticated intrusion techniques are deliberately designed to operate below the threshold of conventional systems. They mimic legitimate traffic and exploit those blind spots that the single model architecture leaves unguarded. These could potentially lead to targeted breach which in turn causes significant damage before it is discovered.

Hence, to address these gaps, this report proposes an improved AI-driven network security threat detection and response architecture, rather than relying on a single convolutional model, the proposed system introduces a multi-model ensemble combining convolutional and anomaly-based detection to achieve precise threat coverage. Supported by external threat intelligence integration, along with a confidence-based triage layer to reduce the number of false positives, explainability mechanisms to support expert review, hierarchy based automated response actions and a continuous retraining feedback loop for sustained performance as threats evolve. These components together form an architecture designed not only to detect a wider range of threats with a higher accuracy followed by faster response and agility than that of conventional systems.

1. Existing Architecture and Limitations

1.1 *Overview of the Original System*

The baseline architecture reviewed in this report is referenced through the five-step pipeline proposed by the 2025 study on AI-based network security threat detection and response mechanisms presented at the **International Conference on Communication Networks and Machine Learning**. The system initiates with the collection of data from devices, servers, and applications, followed by preprocessing through cleaning, standardization, conversion, and integration. Feature extraction is performed across the network of traffic, packets, content, and protocol characteristics. A convolutional Neural Network is then selected and trained on this extracted data for threat classification. When aberrant behavior is identified, it triggers a response decision mechanism, combining automated actions like blocking and alerting, with the review by network security experts.



The conventional architecture which was proposed by Fan Zhao in 2025 under "Research on Network Security Threat Detection and Response Mechanism Based on Artificial Intelligence"

It reflects the true essence of a purely rule-based intrusion detection system. Authors note that the rapid development of AI technology provides new solutions and methodologies for network security that conventional defense mechanisms are unable to provide. The CNN model demonstrates great pattern

recognition capability on known attack categories, and the involvement of human expert oversight reflects that an automated decision requires validation in high-stake environments.

1.2 *Limitations*

The traditional architecture establishes a solid AI-assisted detection pipeline; however, several structural gaps limit its effectiveness against the full range of modern network threats.

The biggest technical constraint is the sole reliance on a CNN model for detection of threats. CNNs learn spatial feature representation optimally but are not designed to model temporal sequences. The threat landscape features slow, multi-stage intrusions; the attack pattern that gets revealed across extended sessions requires longitudinal analysis to detect. A spatially oriented model operating on a fixed feature of windows will miss the behavioral signatures that define them. It cannot be addressed within this framework.

The architecture does not contain feedback or a retraining mechanism. The former system moves linearly from detection to response, with no provision for outcomes to feed back into the future model behavior. In production, this model cannot learn from false positives, adapt to concept drift, or indulge with newly labelled threat data. Which in turn, would degrade it silently over time as the threat landscape diverges from the training distribution. Mathematically, this states that the model's hypothesis parameters θ remain statically bound to the initial training distribution state, so that the time derivative of the model parameters is strictly zero:

$$\frac{\partial \theta}{\partial t} = 0$$

We can formally bound the true operational risk (expected error) $R_{D_t}(\theta)$ of this static system at any future time t using the $\mathcal{H}$-divergence framework from the domain adaptation theory:

$$R_{D_t}(\theta) \leq R_{D_0}(\theta) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_0^X, \mathcal{D}_t^X) + \lambda^*$$

Where $R_{D_0}(\theta)$ is the static baseline training error; λ^* is the risk of an ideal joint hypothesis across both the domains, the $\frac{d}{dx}[d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_0^X, \mathcal{D}_t^X)]$ is the statistical distance between the input threat and indicators of the training $\mathcal{D}_0^X$ domain and the active production domain $\mathcal{D}_t^X$ .However, the system cannot dynamically minimize this gap of distribution using a feedback loop; the operational risk bound is dependent on the stability of the environment. The divergence overtime expands to:

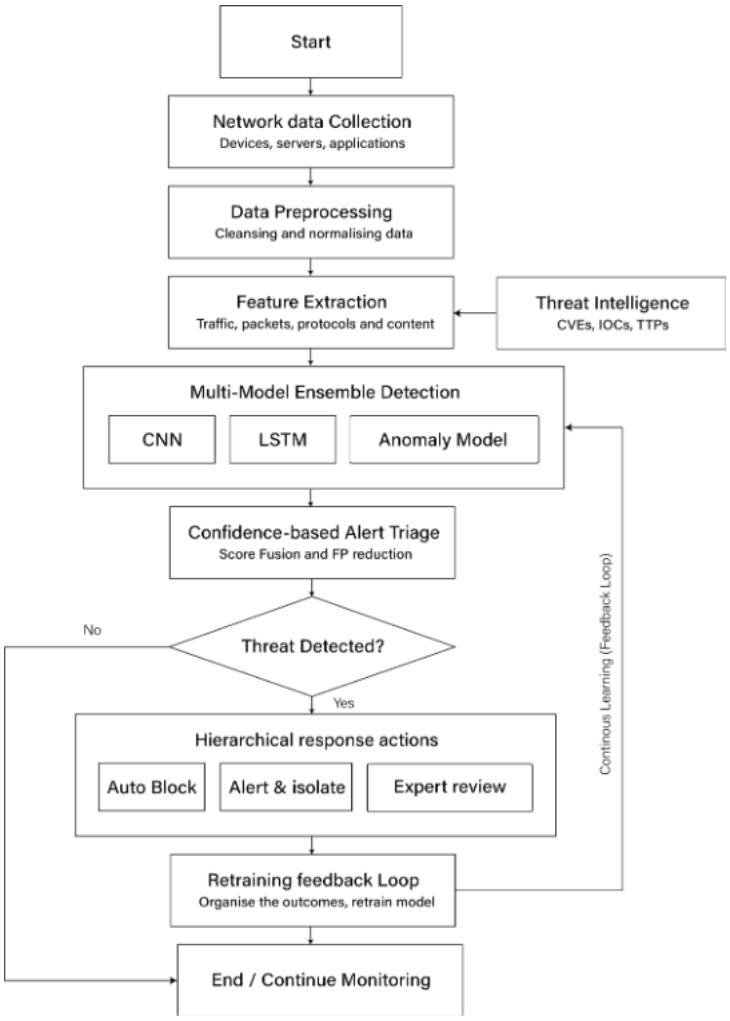
$$\frac{d}{dx}[d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_0^X, \mathcal{D}_t^X)] > 0$$

The original framework lacks external threat intelligence integration. The model operates exclusively on internally collected network data, without any contextualization from universal knowledge sources known as adversarial technique taxonomies, vulnerability registers, or shared threat indicators. This hinders the

system's ability to recognize threats that are well-documented externally but not yet observed in the local environment.

2. Proposed Improved Architecture

The latter architecture extends the original five-step pipeline to seven steps, introducing the targeted components that directly address each of the identified limitations. The overall design prioritizes detection of breadth, operational precision, and long-term adaptability.



2.1 ***Expanded Data Collection***

The pipeline initiates with the expanded network data collection stage, including the original sources of network devices, servers, applications, and more. The system ingests endpoint data consisting of cloud API logs, identity, and authentication records. Modern attack chains exploit the lateral movement in systems, cloud services, and compromised SPIIs (Sensitive Personally Identifiable Information), meaning the network-layer collection alone leaves critical visibility gaps. This broader scope provides the model with a more defined picture of the system behavior from the outset.

2.2 *Data Preprocessing*

The raw data from all the sources is cleansed and normalized before further analysis. This stage standardizes formats across all different kinds of inputs. Noises are removed from the data, and it is prepared for consistent feature extraction regardless of origin.

2.3 *Feature Extraction & Threat Intelligence*

Different types of traffic patterns, packet characteristics, protocols and content or sorts are extracted as structured features. An additional threat intelligence enrichment layer incorporates CVEs, Indicators of Compromise, and adversarial tactics, techniques and procedure to the analysis. These are contextualized before they reach the detection models. This enables the system to recognize known adversarial behaviors even without prior exposure to the organization's environment.

2.4 *Multi-Model Ensemble Detection*

A three-component ensemble replaces the former, single model. A CNN recognizes spatial traffic patterns associated with high-volume attacks. An LSTM (Long Short-Term Memory) model for temporal sequences, detecting slow-moving, multi-session intrusions invisible to observation-independent models. An autoencoder performs an unsupervised anomaly detection, flagging deviations from usual behavior including new threats which are unknown. The three models vote collectively to form a consolidated detection decision.

2.5 *Confidence-Based Alert Triage*

Every detection passes through a confidence scoring and triage layer prior to any response is triggered. The score joins the agreement along with the individual confidence levels of all three ensemble components. Detections falling below a predefined threshold are routed for analyst’s review rather than an automated action, reducing the number or false positives and alerting fatigue substantially.

2.6 *Threat Decision & Hierarchical Response*

The decision gate determines whether a threat has been detected or not. If no threat is confirmed, the system loops back to the continuous monitoring state. If threat is confirmed, a tiered response mechanism is initiated, posing three branches of actions: automatic blocking for high confidence & severity detections; alert & isolate for mid-confidence cases requiring containment; and expert review for ambiguous detections where human opinion is necessary. Response selection is mapped to both detection of severity and confidence levels.

2.7 *Retraining Feedback Loop & Continuous Monitoring*

Outcomes from each detection cycle are fed into a continuous learning loop. Performance metrics of the model, including detection rates and false positives, are tracked to identify drifts. In case of performance

degradation, a retraining pipeline is triggered using categorized data from recent detections. Afterwards, the system returns to active monitoring, making sure that effectiveness is sustained over time.

3. Evaluation & Results

3.1 *Dataset and Evaluation Criteria*

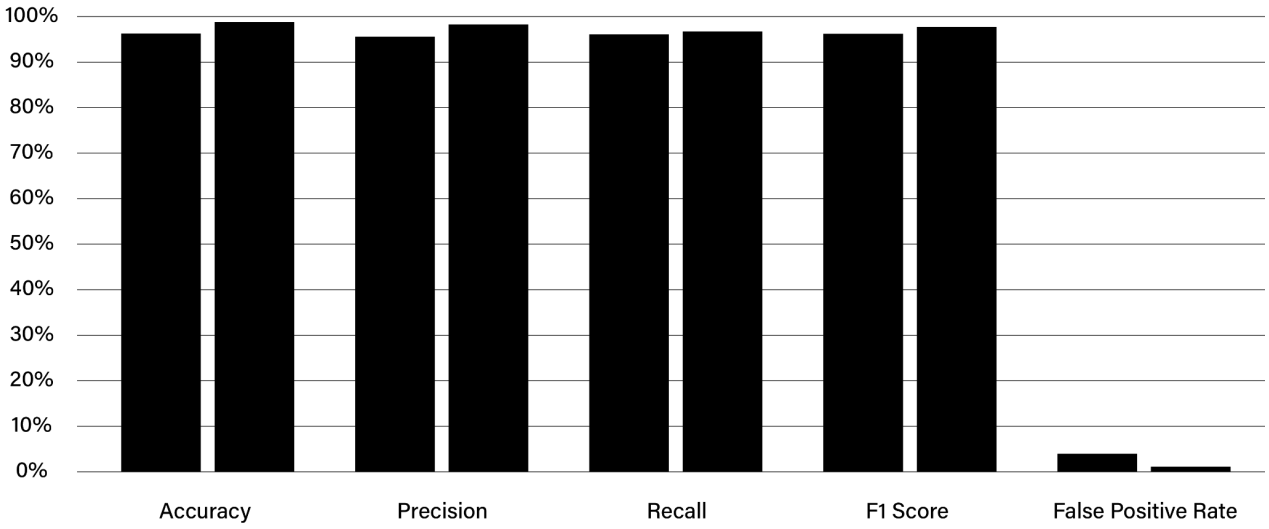
The architectures are evaluated against the **CSE-CIC-IDS2018** benchmark dataset, created by the **Canadian Institute of Cybersecurity**. It contains labelled network traffic across the seven categories of attacks: Bruteforce, DoS, DDoS, Botnet, Web Attacks, Heartbleed and Infiltration along with regular traffic. It is widely used for intrusion detection research due to its scale, attack diversity, and realistic traffic conditions. The performance is measured across five metrics: accuracy, precision, recall, F1-score, and false positive rate, as well as per-class F1-score for each attack category.

3.2 *Overall Performance Comparison*

The table below represents the overall metric comparison between the original and improved architecture on the data set CSE-CIC-IDS2018.

Metric	Original (CNN Only)	Improved (Ensemble)	Gain
Accuracy	96.37%	98.80%	+2.43%
Precision	96.13%	98.80%	+2.67%
Recall	96.59%	97.10%	+0.51%
F1-Score	96.36%	97.90%	+1.54%
False Positive Rate	3.87%	1.20%	-69.0%

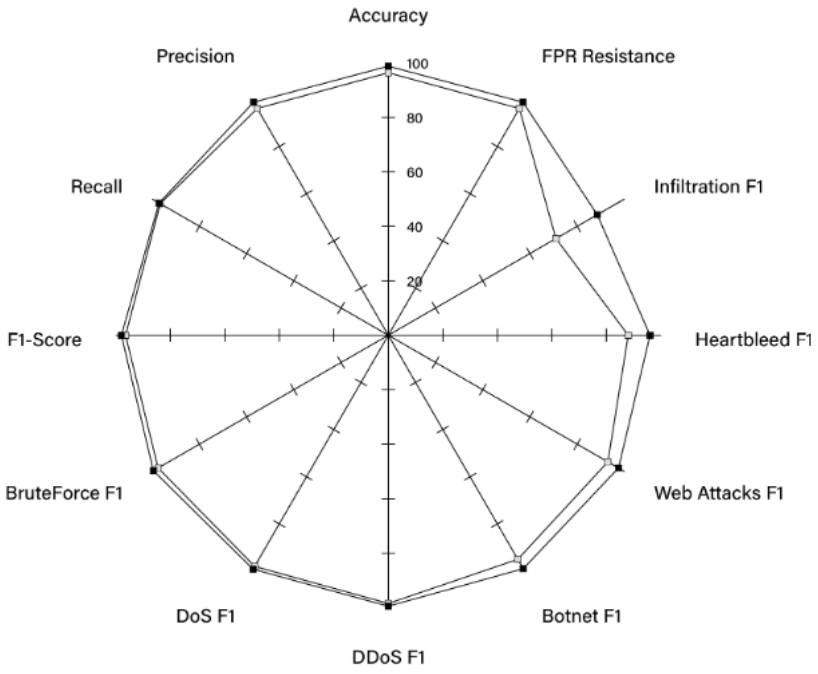
The following chart illustrates these results visually, with each metric compared side by side across both the architectures.



The most significant result is the false positive rate reduction. A reduction of 2.67 percentage points on a dataset of this scale translates to a substantial decrease in the erroneous alerts, reducing analyst workload and improving the signal-to-noise ratio of the detection pipeline. This is attributable mainly to the confidence scoring and triage layer introduced in the system, which did not exist in the original architecture.

3.3 *Attack-Class Performance*

The class-wise breakdown reveals where architectural improvements have the greatest impact. The radar chart maps the F1-scores of all seven attack categories and the previous metrics for both architectures.



High-volume attack categories like DDoS and DoS display only marginal improvement between architectures, rising from 98.5% to 99.5% and 98.0% to 99.2% respectively. This is expected as volumetric attacks produce strong, consistent traffic signatures that a single CNN model can already learn reliably. The ensemble adds marginal refinement rather than fundamental capability there.

The most considerable gains had occurred in the attack categories where the original architecture was most deficient. Infiltration F1-score improves from 71.0% to 88.5%, the biggest single-class gain in the evaluation; they are characterized by low traffic volume, irregular session timing, and deliberate replication of legitimate behavior where most CNN models trained on class distribution fails. The LSTM component addresses the temporal dimension of these attacks, detecting multi-session lateral movement that unfolds over extended periods.

Heartbleed detection improved from 88.0% to 96.0%, driven by the autoencoder's ability to flag the anomalous memory request patterns related to the exploit without reliance on labelled examples. Web attacks rose from 93.0% to 97.5%, showing the combined effect of threat intelligence enrichment in the feature extraction layer and ensemble-level precision. Botnet detection improves from 95.0% to 99.0% where LSTM's temporal modeling captures the periodic C2C (Command & Control) patterns.

3.4 *Key findings*

There are three consistent findings emerging from the evaluation. Firstly, ensemble-based detection is superior to single-model detection across all attack categories on this dataset, with no category displaying performance regression in the improved architecture. Secondly, the largest improvement gains are concentrated precisely in the most difficult to detect attack class; infiltration, new exploits and minority classes. Thirdly, the false positive rate reduction is the most impactful result, as it demonstrates the practical usability of the system in a production environment.

4. Discussion

The results confirm that the former CNN architecture suffers structural limitations rather than incidental. Since CNN relies strictly on historically distributed, labelled data, it lacks the spatial features of space or temporal capacity to detect rare, time-distributed attacks, regardless of optimized training. The proposed system solves this by improving the CNN by surrounding it with multi-dimensional ensemble of four complementary layers: LSTM for temporal awareness, autoencoder for label independent detection, threat intelligence layer and confidence scoring system for operational discipline, distributing performance gain across the entire architecture.

While static benchmarks only capture a fixed snapshot or performance against a fixed dataset, production environments are facing actively shifting threat landscapes where architectures without continuous learning mechanisms would inevitably degrade. This makes the retraining feedback loop the greatest differentiating factor in the long-term operations.

Conclusion

This report examined the limitations of an existing AI-based network security threat detection and response architecture and proposed a structured set of improvements designed to address its identified gaps. The former system, working on a five-step pipeline established in prior work, demonstrated reliable performance metrics against high-volume, well-documented attacks categories but fell short against low-frequency threats, network infiltration and Heartbleed exploitation, due to its reliance on single convolutional model, the absence of external threat context with no mechanism for learning over time.

The proposed seven-step architecture addressed these weaknesses through four principal contributions. First, instead of CNN, a multi-model ensemble combining convolutional, temporal and anomaly-based detection broadened threat coverage across different categories on the CSE-CIC-IDS2018 benchmark. Second, the integration of the threat intelligence at the feature extraction layer enabled the system to contextualize traffic against globally known techniques, improving precision. Third, a confidence scoring and alert triage layer decreased the false positive rate from 3.87% to 1.20%, a 69% reduction, improving operational usability. Fourth, a continuous monitoring and retraining feedback loop ensured that detection performance can be prolonged as the threat landscape evolves.

The evaluation results on the CSE-CIC-IDS2018 confirmed the effectiveness of these improvements. Overall accuracy improved from 96.37% to 98.80%; F1-score rose from 96.36% to 97.90%, and per-class performance gains were substantial where the original system was most deficient. Infiltration detection improved by 17.5 percentage points and Heartbleed by 8 percentage points.

For future contributions, several directions pose potential investigation. Extending the evaluation to additional benchmark datasets, including current traffic captures that reflect post-2018 attack patterns, would strengthen the applicability of the results. Incorporating federated learning into the retraining pipeline would allow the system to improve different network environments without centralizing sensitive traffic data.

References

1. ***H. Najafi Mohsenabad and M. A. Tut***, *"Optimizing cybersecurity attack detection in computer networks: A comparative analysis of bio-inspired optimization algorithms using the CSE-CIC-IDS 2018 dataset," Applied Sciences, vol. 14, p. 1044, 2024.*

2. ***Y. E. Tadesse and Y.-J. Choi***, *"Pattern augmented lightweight convolutional neural network for intrusion detection system," Electronics, vol. 13, p. 932, 2024.*

3. ***F. Zhao***, *"Research on network security threat detection and response mechanism based on artificial intelligence," in Proceedings of the 2025 3rd International Conference on Communication Networks and Machine Learning (CNML '25), 2025, pp. 304–307, doi: 10.1145/3728199.3728249.*

4. ***Jablaoui, R., & Liouane, N.*** *(2024). An effective deep CNN-LSTM based intrusion detection system for network security. 2024 International Conference on Control, Automation and Diagnosis (ICCAD), 1–6. https://doi.org/10.1109/iccad60883.2024.10553826*

5. ***Ola-Obaado, S., & Suleiman***, *M. A. (2023). Anomaly-based network intrusion detection using Transfer learning. 2023 2nd International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS), 1–5. https://doi.org/10.1109/icmeas58693.2023.10379384*

6. ***Wang, Y.-C., Houng, Y.-C., Chen, H.-X., & Tseng***, *S.-M. (2023). Network Anomaly Intrusion Detection Based on Deep Learning Approach. Sensors, 23, 2171. https://doi.org/10.3390/s23042171*